

EREBUS OBSERVATORY NETWORK / EON

Tomorrow's Newspaper

Testing a language-model analyst on filings it could not have read

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Technical white paper

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A language model read 6,653 annual reports. This is how to test it honestly.

Research, not investment advice.

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Abstract

EON began in May 2025 as a folder of scripts that read 11,741 annual reports to learn what excellent companies share. It became a system that reads every 10-K of the listed market through three fixed lenses and ends each reading in one verdict: 6,653 company-years of 1,358 companies, 500 a day, at the ceiling of a free-tier quota.

An early backtest looked excellent. But most of its outcomes ended before the model's knowledge cutoff of January 2025, so the model may simply have known how those years ended. **A language model's verdicts can only be tested on filings it could not have read.**

On a protocol fixed in advance, the one-year BUY-minus-SELL spread before the cutoff is 13.1 points ($p = 0.0001$), and about 40% of it is an industry tilt. After the cutoff, the tilt is gone; a rank spread of 5.9 percentile points remains ($p = 0.0051$), on one vintage and a statistic added late.

The 2025 scripts left something better: scores recorded before their outcomes existed. Tested now, the headline score, resemblance to 39 excellent companies, pointed the wrong way (Spearman -0.14 , $p = 0.0001$). A contrarian score and options calls showed nothing. The clean tests do not agree, and that is the finding: a fluent reader is not thereby a forecaster. Fix the model, date every reading, and keep the ledger.

This is research, not investment advice.

1. The question

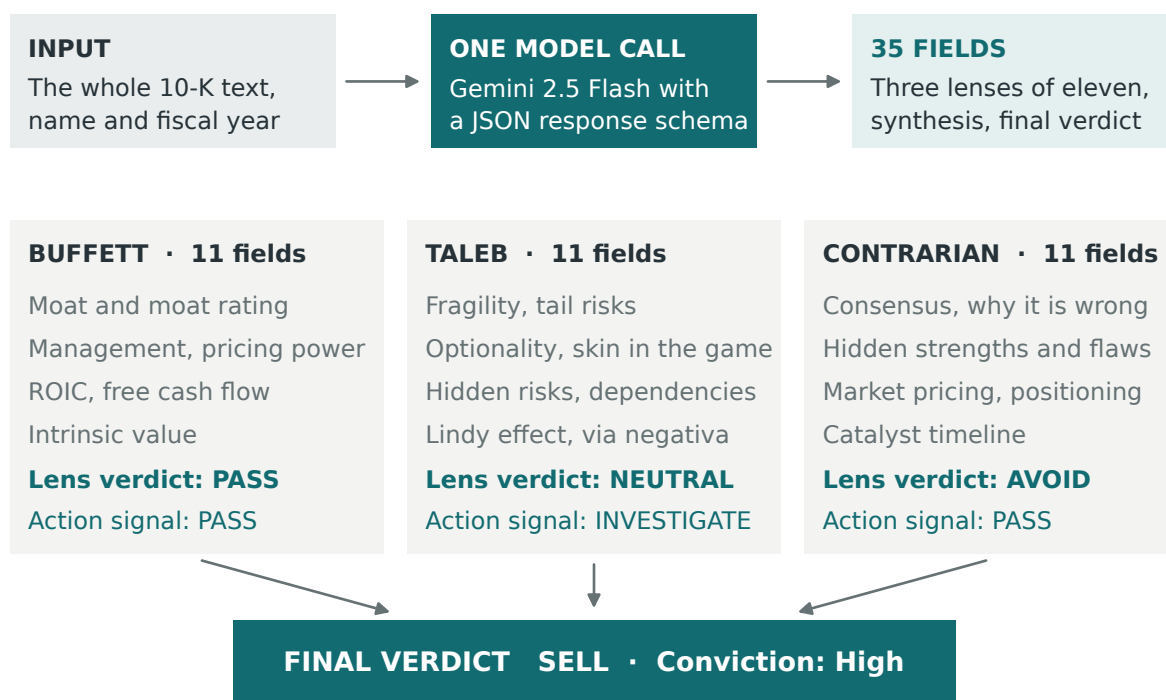
1.1 The interaction

A 10-K is the most complete account a public company gives of itself. It is also long, repetitive and read closely by few people. I wanted a reader that never tires: one that would give every filing the same careful attention and leave a record I could count.

EON gives a model one filing and asks one fixed question in three parts. As Warren Buffett would: is there a durable moat, a capable management, a fair price? As Nassim Taleb would: what breaks this company, and what would make it stronger under stress? As a contrarian would: what does the consensus believe, and why might it be wrong? A synthesis weighs the three and ends in a verdict: STRONG BUY, BUY, HOLD, SELL or STRONG SELL, with a conviction level.

One filing becomes one countable verdict

Specimen: Apple 10-K for fiscal 2025, filed 31 Oct 2025 / illustrative, not evidence



The backtest reads only the first word: STRONG BUY, BUY, HOLD, SELL or STRONG SELL.

Source: data/eon.db, stored reading 1. Filed after the model's cutoff; no outcome is claimed.

Figure 1: One filing becomes one countable verdict. The specimen is the first stored reading: Apple's 10-K for fiscal 2025, filed 31 October 2025. The three lens verdicts disagree in wording; the synthesis resolves them into one. The filing postdates the model's knowledge cutoff, and no outcome is claimed for it. Illustrative, not evidence. Source: data/eon.db.

The readings are specific. Apple's Taleb lens finds "Total Debt/EBITDA ratio of 0.68x in 2025 (\$98,657M debt / \$144,748M EBITDA)", well below its 3x concern threshold. The final verdict begins: "SELL. Conviction: High. Apple is an undeniably excellent company, but a

high conviction SELL rating is warranted given the significant disconnect between its robust fundamentals and its highly stretched valuation.”

The verdict is what makes a reading countable (Figure 1). The prose around it is for a person; the first word is for measurement.

1.2 The constraints

Constraint	Why it binds
Free-tier model quotas	25 keys at 20 requests a day: 500 readings a day, at most
The whole filing, every time	No summaries or excerpts; the model reads what an analyst would
A fixed schema	Readings must be comparable across companies and years
Every result survives a restart	A market-wide batch runs for weeks and will be interrupted
An evaluation that can fail	A test the model can pass from memory measures nothing

The first and last constraints shape the paper. The quota decided how EON is built. The last constraint decided how it is judged.

1.3 Which evidence belongs to which claim

Several samples appear below. They overlap, so each claim is tagged against this table.

Sample	Size	What it is	Used for
All readings	6,653	Unique company-years from the three-lens schema, 1,358 companies	Verdict mix (§6)
February backtest	4,689	Readings available on 12 February 2026, 515 tickers	Reported results (§7.1)
Priced readings	6,203	Readings with a parsed verdict, a filing date and prices	This evaluation (§9)
Before the cutoff	1,099 BUY, 657 SELL	One-year window closed before 31 January 2025	Historical test (§9.1–9.2)
After the cutoff	339 BUY, 201 SELL	Filed after 31 January 2025, one-year outcome observed	Clean test (§9.3)
2025 ledger	1,784 / 1,797 / 874	Compounder, contrarian and options scores from May–June 2025, with outcomes	Forward test (§9.5)

2. Where it started: a study of excellence

Before EON had a name, it was a folder of Python scripts asking a narrower question: what do excellent companies have in common, and who else has it?

In early May 2025 the scripts downloaded up to thirty years of 10-Ks for 39 long-run compounders and had Gemini 2.5 Flash, then an April preview, read each one. The readings were distilled into a meta-analysis of shared success factors, dated 9 May. Between 10 and 13 May the same scripts read the recent filings of about 1,900 other companies and scored each from 0 to 100 on its resemblance to that pattern. In all: 11,741 filing readings of 2,175 companies, covering fiscal years 1994 to 2025, as many as 4,641 in a single day.

Two more passes followed. On 26 May a contrarian scan scored 1,933 companies on six “alpha” dimensions. On 4 June an options pass read the same companies again and made 951 directional calls: buy calls or buy puts.

Almost everything in EON descends from these weeks: reading whole filings, the contrarian scanner’s six dimensions, the options workflows, and the corpus of excellent-company factors that the Moonshot × Excellence screener still loads.

So do two of its lessons. The outputs had no fixed schema. The script that later flattened them found the six success factors spelled about 130 different ways. And the model did not know what day it was. Run in May 2025, it stamped 1,916 of its 1,919 comparative analyses with a date in 2024, and 91 of the options readings suggested expiries in 2024, already in the past.

The 2025 outputs were saved and never changed. That turned out to be their most valuable property: they are a ledger written before its outcomes (\$9.5).

3. Design one: a workflow builder

The project, then called Fintel, returned in December 2025 as an application. Its centrepiece was a visual workflow builder. A user chained steps (choose companies, fetch filings, run an analysis, filter, aggregate, export) and ran the chain. The archive holds 16 saved workflows and 26 workflow runs from 7 to 30 December 2025, among 250 analysis runs by mid-January.

It was the right way to learn which questions were worth asking. The saved chains compare three chip makers, read three nuclear start-ups through one lens, and pull executive pay from a proxy statement.

It was the wrong way to learn whether any answer was right. Each chain asked its own question of its own companies. No two runs were comparable, and nothing accumulated. A pile of good essays is not a dataset.

The builder was removed on 30 December 2025. What it taught stayed: the three lenses, the structured output, and the idea, first glimpsed in May, that the question should be written once and asked of everything.

4. Design two: one question, every filing

4.1 The schema is the instrument

Design two fixes the question. One prompt, one response schema, and every listed company’s recent 10-Ks. The schema has 35 fields: eleven for each lens, a synthesis, and the final verdict. Pydantic validates every response; a reading that does not fit the schema is not stored.

Fixing the question turns a model’s opinions into a panel: the same measurement, taken on 1,358 companies over six fiscal years. That is what makes everything later in this paper possible.

A fixed question asked of every filing turns a language model from a conversation into an instrument.

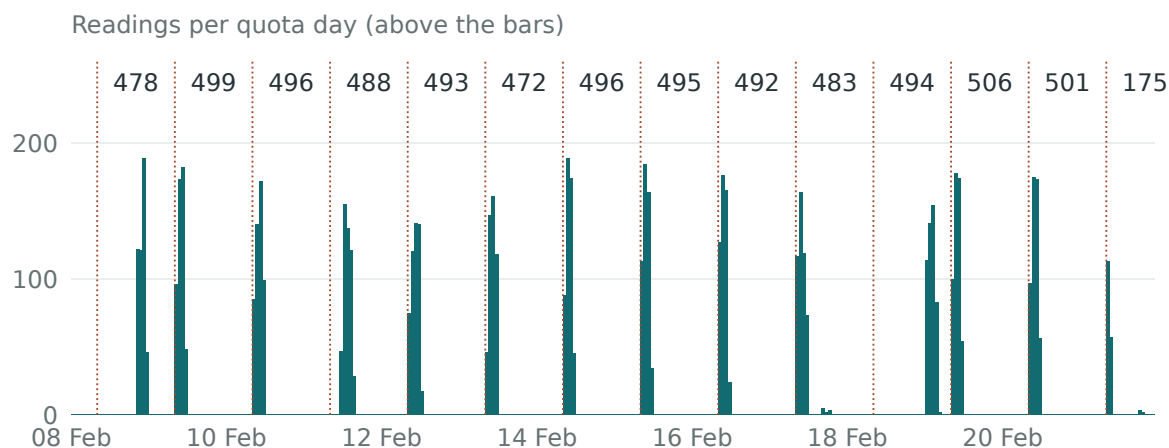
4.2 The quota is the clock

Every reading is one request, and requests are rationed. EON rotates 25 API keys, each allowed 20 requests a day, with file locks so that parallel workers, the command line and the web interface never spend the same request twice.

The market-wide batch ran from 8 to 21 February 2026. It read 6,568 filings from 1,327 companies. Figure 2 shows its rhythm: a burst after the quota reset at midnight Pacific, then silence until the next.

The quota is the clock

Market-wide batch, 8 to 21 February 2026 / readings per hour, UTC



25 keys × 20 requests = 500 readings a day. Dotted lines: quota reset, midnight Pacific.

Source: data/eon.db, batch all_comp_08022026. The last quota day is partial.

Figure 2: The quota is the clock. Every full quota day read between 472 and 506 filings, against a ceiling of 500; the day boundary here is approximate, so a day can exceed it slightly. The last day is partial. Hours are UTC; the reset is midnight Pacific, 08:00 UTC in February. Source: data/eon.db, batch all_comp_08022026.

A clock this slow changes what engineering matters. A 1,000-company, ten-year batch is 20 days of quota. So EON saves each fiscal year's result the moment it arrives, holds a lease on each company with a heartbeat, and resumes an interrupted company from its last completed year. When every key is spent, it waits for the reset and continues. None of this is novel. It is what a rationed reader needs.

4.3 What happens to a filing

EON downloads each 10-K's HTML from SEC EDGAR, renders it to PDF in headless Chrome, extracts the text with PyPDF2, and appends all of it to the prompt. No filing is truncated. A filing too long for the model's context is marked skipped, not cut.

The route through PDF is a detour, and it costs a browser process per filing. It does not affect the evaluation, but a cleaner pipeline would parse the HTML directly.

Of the 1,410 companies in the market-wide batch, 83 failed: 61 because no 10-K could be retrieved, and 22 because the model call failed after retries.

5. Design three: workflows as hypotheses

With the panel built, I began writing narrower questions as custom workflows: auto-discovered Python files, each with its own prompt and schema. Two lessons from this period shaped the

tools, and one of them reappears in the results.

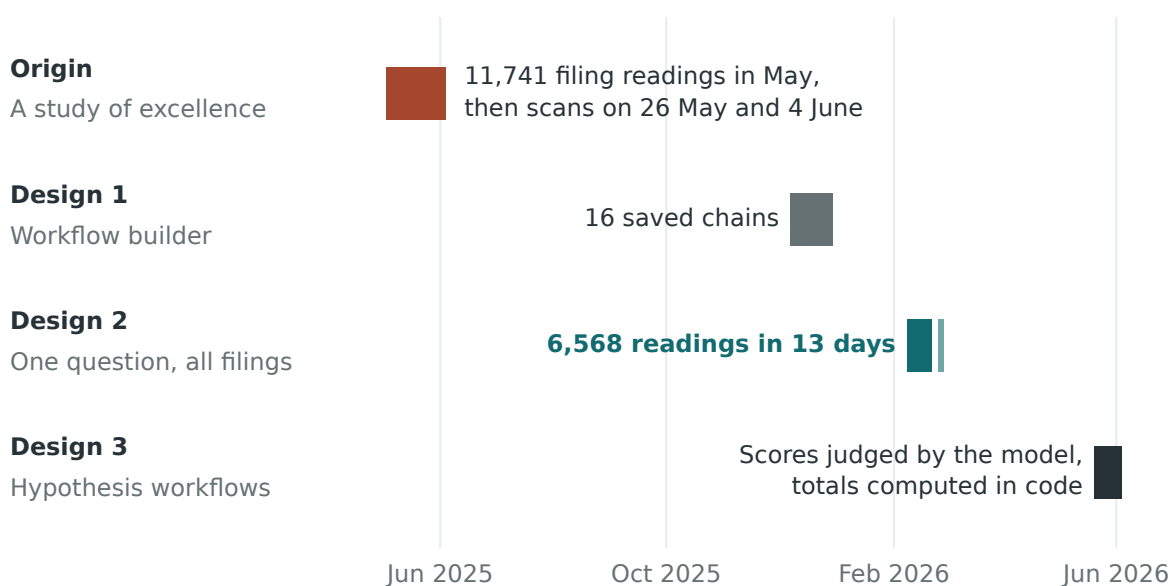
Schema size is a reliability budget. The CSPP workflow first asked for a whole analytical framework in one call. Its schema was about 106 KB, 28 times the size of one lens. When any of its roughly 200 required fields drifted, validation failed and the whole reading was lost. On 20 May 2026 it was split into four calls of lens-sized schemas, merged afterwards, with a partial result kept if one call fails.

Let the model judge; let code count. The same rewrite stopped trusting the model's arithmetic: domain totals and the master score are recomputed in Python from the model's component scores. The Moonshot × Excellence screener of 4 June 2026 goes further. The model rates six dimensions on anchored 0 to 10 scales, each defined at 0, 5 and 10, and code computes the weighted totals.

That screener also shows the quota as a design constraint. It deliberately uses one call rather than two. It injects a reference corpus of up to 250,000 tokens, and a second call would resend it and spend a second request per company.

Fix the question, then read everything

Four designs, May 2025 to June 2026 / dates from files, git history and batch records



Sources: origin files; git log; fintel.db; eon.db. Rust: a ledger written before its outcomes.

Figure 3: Four designs, one idea: fix the question, then read everything. The origin's dates come from its output files, later dates from git history and stored readings. The rust bar marks the 2025 readings, recorded before any of their outcomes existed. Bar length is calendar time, not effort. Sources: the origin project; git log; data/archive/fintel.db; data/eon.db.

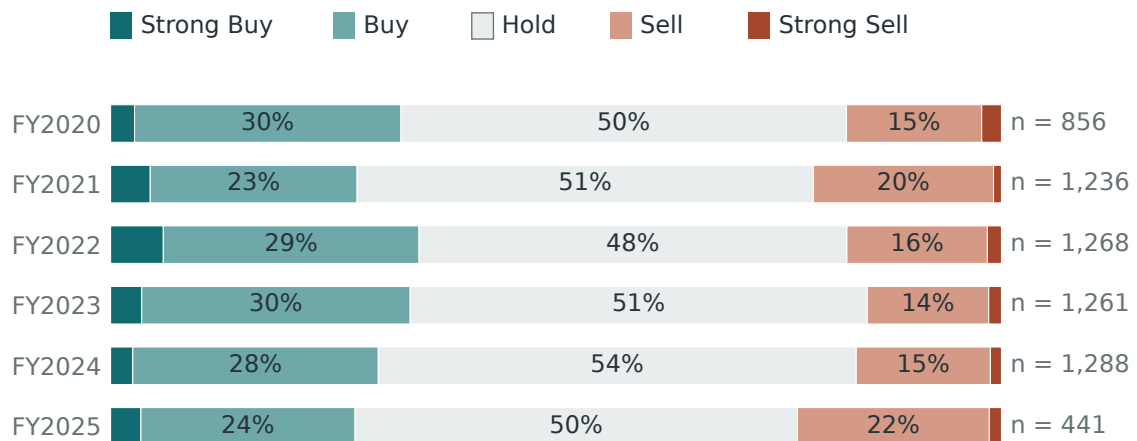
Each design kept what the last one taught (Figure 3). The project has run for seventeen months. None of its designs, by itself, said whether the readings were right.

6. What 6,653 readings said

Before asking whether the verdicts were right, it is worth seeing what they were. Across 6,411 dated readings, 50.8% say HOLD, 27.5% BUY, 16.5% SELL, 3.8% STRONG BUY and 1.4% STRONG SELL. Conviction is Medium in 4,632 readings, High in 1,516 and Low in 148.

Half of every vintage is HOLD

Final verdicts by fiscal-year vintage / vintages with at least 400 dated readings



Source: evaluation/results.json. Vintage = fiscal-year label. Shares under 6% are unlabelled.

Figure 4: Half of every vintage is HOLD: between 48.0% and 53.6% in each fiscal year from 2020 to 2025. SELL and STRONG SELL together range from 15.1% to 22.9%. Vintage is the fiscal-year label EON received from EDGAR metadata. FY2025 is partly covered. Source: evaluation/results.json.

The mix barely moves from year to year (Figure 4), across a speculative boom, a bear market and a recovery. The model is not reading the market's mood into its verdicts, or at least not in its overall proportions. Whether it reads anything useful about individual companies is the next question.

7. The February backtest, and the problem of memory

7.1 What it reported

On 12 February 2026, while the market-wide batch was still running, I ran a backtester over the readings stored so far. It used 4,689 readings of 515 tickers, entered each position on 1 April after the fiscal year, and compared returns with SPY. Its report found that BUY-rated stocks beat SELL-rated ones by 7.1 points over one year ($p = 0.003$) and 14.1 points over two ($p = 0.0008$). The two-year result survived clustering by ticker ($p = 0.041$). High-conviction calls appeared stronger again: 25.5 points over two years. It concluded that the model showed "genuine stock selection skill".

These are reported results from `experimental/backtester/BACKTEST_REPORT.md`, not new measurements.

7.2 The model has read the answers

The report tested every reading against an outcome that had already happened. Gemini 2.5 Flash has a published knowledge cutoff of January 2025. A reading of a fiscal 2021 10-K, entered in spring 2022 and judged in spring 2023, asks the model about a period its training data may describe in detail.

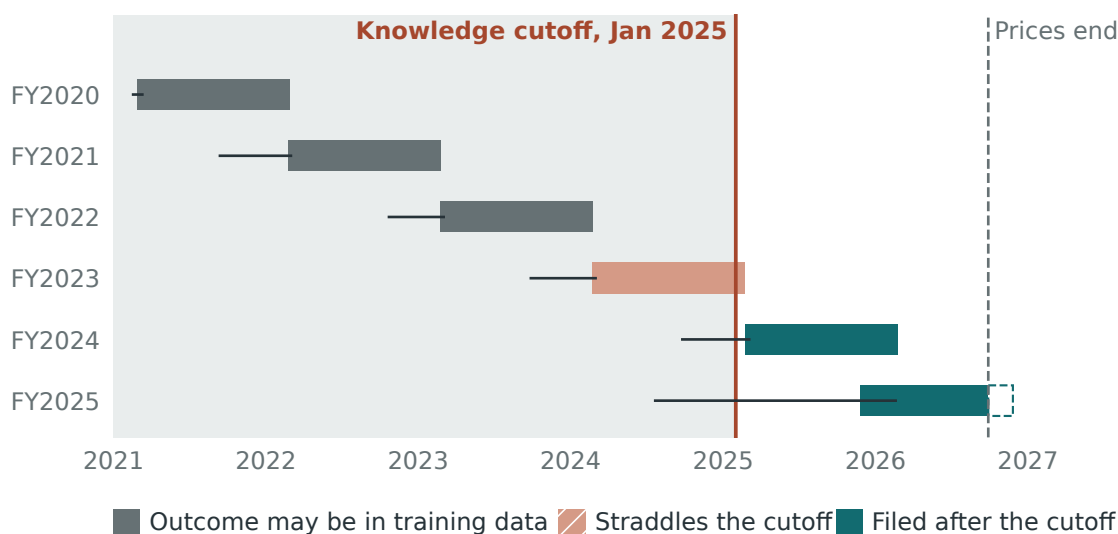
The filing text does not reveal the future. But the prompt names the company and asks for the consensus view, the market's positioning and the catalysts ahead. The verdicts also weigh valuation, and a 10-K states no current share price, only the value of the public float some months before. The model supplies the rest from what it knows. Answering those well invites the model to draw on everything it knows about the company, and it cannot separate what it knew in 2022 from what it learned later.

If a model's training data overlaps the outcome period, a backtest measures its memory as well as its judgement, and cannot say how much of each.

Figure 5 shows how much of the history sits on the wrong side of that line.

Most outcomes predate the model's cutoff

One-year windows by vintage, from median entry / whiskers: 10th to 90th percentile entry



Source: evaluation/trades.csv. Dashed outline: part of the window not yet observed.

Figure 5: Most outcomes predate the model's cutoff. Bars are one-year windows from each vintage's median entry date; whiskers span the 10th to 90th percentile of entries. FY2025's early entries come from companies whose fiscal-year labels run ahead of the calendar; the evaluation classifies every reading by its actual filing date. Cutoff: Google's Gemini 2.5 Flash model documentation, "January 2025". Source: evaluation/trades.csv.

7.3 Three smaller problems

Readings are not independent. A company appears in up to six vintages, and all readings in a vintage share one market. A t-test over trades treats 1,700 correlated bets as 1,700 independent

ones. Clustering by ticker helps with the first problem, not the second.

The universe survived. The companies were listed in February 2026. Firms that failed or were acquired before then were never read, and 45 tickers that were taken private or acquired by September 2026 have no prices in the refresh.

Entry dates were a convention. The 1 April entry was never early: 6,386 of 6,411 dated readings were filed on or before it. It was often late, by up to a year for non-December fiscal years. Actual filing dates are better.

None of these is as serious as memory. All of them are fixable.

8. A test built around the cutoff

I wrote the protocol in `evaluation-config.yaml` before the first run. It fixes five choices.

1. **One reading per company-year**, the first stored. Overlapping batches had saved some twice.
2. **Entry on the first trading day after the actual filing date.** Excess return is the stock's return minus SPY's over the same window, at six months, one year and two years (126, 252 and 504 trading days).
3. **Long means BUY or STRONG BUY; short means SELL or STRONG SELL.** The statistic is the difference in mean excess return between them, the spread.
4. **Permutation tests, not t-tests.** Verdicts are shuffled 10,000 times within each vintage, and separately within each vintage-and-industry block. The null hypothesis is that a verdict carries no information beyond its vintage, or its vintage and industry. This respects the shared market of each vintage, which a trade-level t-test ignores.
5. **Two primary questions.** Does the one-year spread survive for windows that closed before the cutoff? Does a six-month spread appear for filings published after it?

The first run used the price cache EON already had, which ended on 6 February 2026. It showed that the historical effect lives at one year and beyond, where the clean sample had no outcomes yet. Prices through 28 September 2026 made the one-year clean test possible. I made three amendments, recorded in the config with their reasons:

Amendment	Why	Free parameters
One-year test after the cutoff	Mirrors the historical test exactly	None
Rank statistic for every primary sample	One stock moved a mean by ~10 points (§9.3)	None; same blocks
Anachronism probe	A direct look for leaked knowledge (§9.4)	Term list and dates

Two parsing corrections followed: verdicts prefixed “Overall investment recommendation:”, and conviction written several ways. They added 328 readings and changed no conclusion. The first run's outputs are kept in `evaluation/v1-feb-cache/`.

9. Results

9.1 Before the cutoff: large, and growing with time

For windows that closed before 31 January 2025, BUY-rated filings beat SELL-rated ones by 13.1 percentage points at one year: 1,099 BUY readings at +2.2 points over SPY against 657 SELL

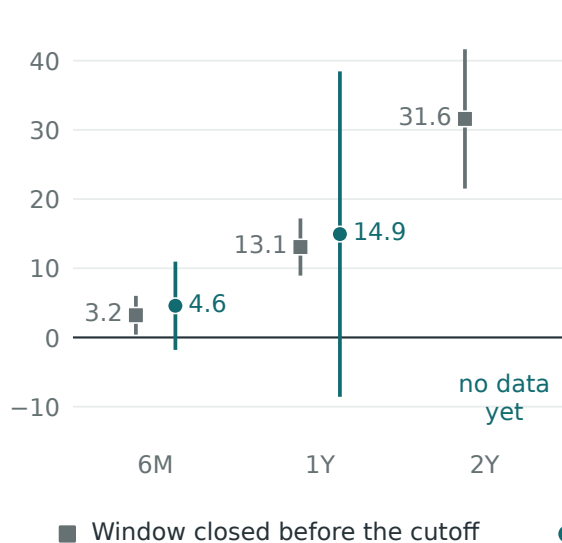
readings at -10.9 . No permutation within vintage, or within vintage and industry, produced a spread as large ($p = 0.0001$ in both, the smallest value 10,000 permutations can give).

The spread grows with the horizon: 3.2 points at six months, 13.1 at one year, 31.6 at two (Figure 6, grey). At one year the short leg carries most of it. At two years both legs contribute: $+17.0$ for BUY, -14.6 for SELL.

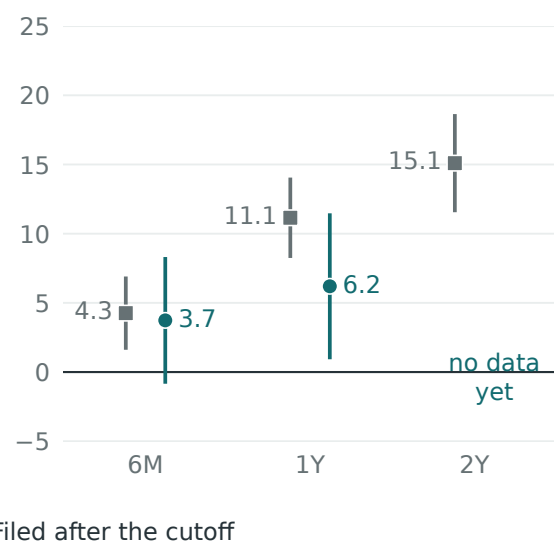
After the cutoff: same sign, wider intervals

BUY minus SELL, excess over SPY / 95% intervals / cutoff 31 January 2025

A Mean spread, percentage points



B Rank spread, percentile points



Source: evaluation/results.json. Intervals: 1.96 Welch standard errors.

Figure 6: After the cutoff: same sign, wider intervals. Grey squares: windows that closed before 31 January 2025. Teal circles: filings published after it. Intervals are 1.96 Welch standard errors, shown for comparison; the tests in the text are permutation tests. Rank spread is the difference in mean within-vintage percentile rank. Two-year outcomes do not yet exist after the cutoff. Source: evaluation/results.json.

The sign holds in every vintage, not just on average:

Vintage	BUY / SELL readings	1Y mean spread	1Y rank spread	Window
FY2020	268 / 144	+18.0	+14.9	Before cutoff
FY2021	331 / 253	+11.7	+8.8	Before cutoff
FY2022	415 / 212	+16.4	+12.1	Before cutoff
FY2023	407 / 182	+8.0	+7.0	Straddles
FY2024	372 / 203	+9.6	+5.3	Mostly after

Spreads are in percentage and percentile points; vintages are fiscal-year labels, so a few FY2024 filings predate the cutoff. The spread is largest for the vintages entered in early 2021 and early 2023. On the rank measure it is less than half as large for the two most recent.

Taken alone, this would support the February report. It is also exactly what memory would produce.

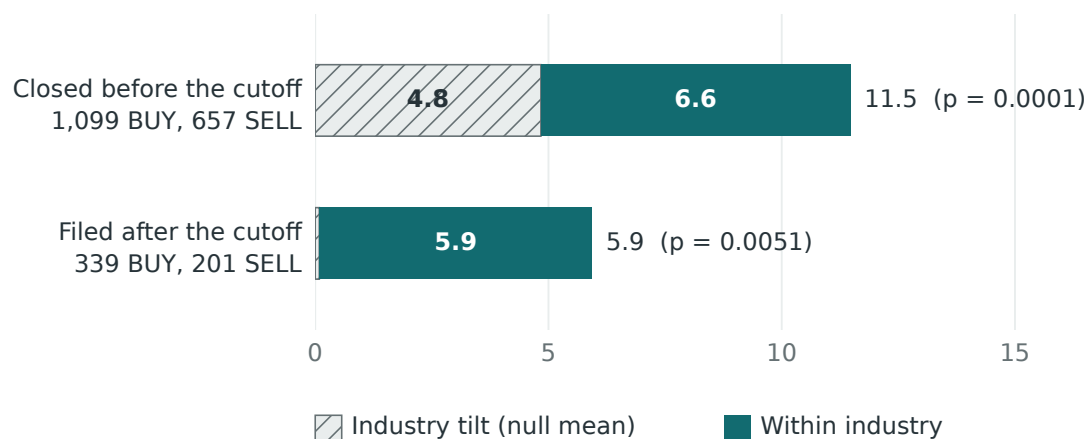
9.2 Where the historical spread comes from

The within-industry permutation test separates two things. The mean of its null distribution is the spread you would expect from the industries the model favoured, with verdicts otherwise random. The remainder is selection among companies in the same industry.

On the rank statistic, the one-year historical spread of 11.5 percentile points splits into 4.8 points of industry tilt and 6.6 within industry (Figure 7; parts are rounded separately). About 40% of the historical effect is a bet on industries, and the industries it favoured went on to do well.

The cutoff removed the industry tilt, not the rest

One-year rank spread, split by a within-industry permutation test / percentile points



Source: evaluation/results.json. Rank statistic added after seeing the tails.

Figure 7: The cutoff removed the industry tilt, not the rest. Industry tilt is the mean of the within-industry permutation null; the remainder is the within-industry spread. Totals (11.5 and 5.9) differ slightly from Figure 6 (11.1 and 6.2) because the permutation test drops readings alone in their industry block. Industries are a February 2026 FactSet snapshot. The rank statistic was added after inspecting the tails (§8). Source: evaluation/results.json.

9.3 After the cutoff: smaller, noisier, not zero

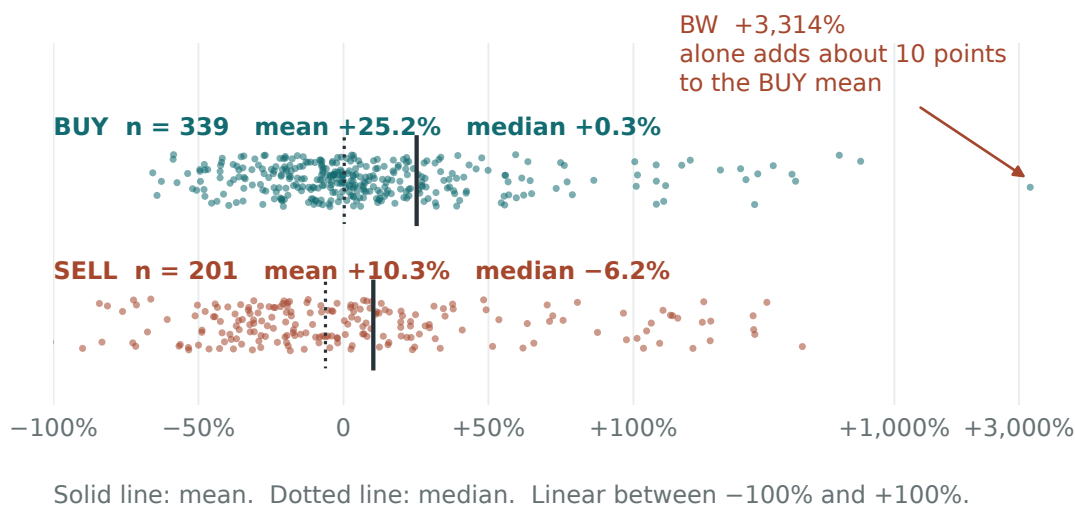
After the cutoff there are 1,443 filings, published between 3 February 2025 and 17 March 2026. Most are fiscal 2024 10-Ks filed in February and March 2025. The model could not have read any of them in training. The readings were made in February 2026 with web search switched off, so the filing itself was the model's only source of anything after January 2025.

At six months (the pre-registered clean test), 416 BUY and 275 SELL readings give a spread of 4.6 points. The within-vintage permutation gives $p = 0.11$; the within-industry test gives $p = 0.048$. The test could only have reliably detected a spread of about 7 points, and the historical six-month spread was 3.2. It neither confirms nor refutes the historical result.

At one year, 339 BUY and 201 SELL readings give a mean spread of 14.9 points, almost exactly the historical 13.1. The interval is ± 23 points and the permutation p-values are 0.22 and 0.29. The reason is in Figure 8.

One stock can decide a mean

Filed after the cutoff / one-year excess return over SPY per reading, symmetric log scale



Source: evaluation/trades.csv. Filed after 31 Jan 2025; outcome by 28 Sep 2026.

Figure 8: One stock can decide a mean. BW's 10-K of 31 March 2025 was read as BUY; the stock returned 3,314% more than SPY over the following year, adding about 10 points to the BUY mean on its own. BUY's mean is +25.2% and its median +0.3%. Returns are Yahoo Finance adjusted closes; BW's has not been checked against a second source. Source: evaluation/trades.csv.

A statistic one company can decide cannot answer the question. The rank spread, added for this reason and applied to every sample alike, gives 5.9 percentile points ($p = 0.0051$), with a detectable size of 6.0. Almost none of it is industry tilt: 0.1 points, against 4.8 before the cutoff. The within-industry part is 5.9, against 6.6 before (Figure 7).

9.4 A direct look for leaked knowledge

If the model's later knowledge leaked into its readings, it might occasionally say so. I searched every reading for terms that entered public use on known dates, and counted those appearing in readings of filings published earlier.

The probe has power for one term. The Inflation Reduction Act appears in 300 readings, and in none of the 2,124 priced readings of filings published before it was signed on 16 August 2022. The other terms are too rare to test anything: the CHIPS and Science Act appears 12 times, the rest at most twice. None appears early.

This rules out careless leakage, where the model writes about events the filing could not contain. It cannot rule out quiet leakage: a verdict nudged by what the model knows, with nothing in the prose to show it.

9.5 A ledger written in 2025

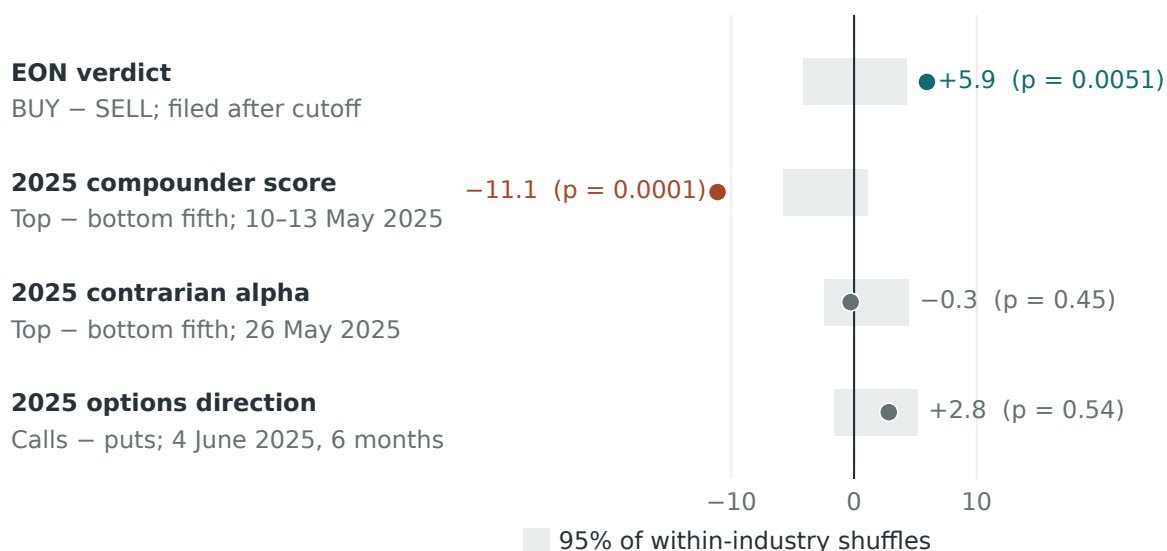
Everything above depends on an argument about what the model could have known. The 2025 scores need no such argument. They were recorded in May and June 2025, after the model's cutoff and before any of the returns they might predict, and left untouched since.

I tested them under a protocol written before the test, with the same rules as the main evaluation: entry on the first trading day after each reading, excess return over SPY for one year (six months for the options calls, their own timeframe), permutations within industry. One earlier look must be disclosed: on 14 July 2026 I checked the compounder score against raw returns and found the top fifth had returned 10.4% against 35.6% for the bottom fifth.

The protocol confirms it. Across 1,766 companies, resemblance to the 39 excellent companies predicted *underperformance*: Spearman -0.14 ($p = 0.0001$). The top fifth trailed SPY by 17.8 points on average and the bottom fifth beat it by 18.6; the medians, -26.5 and -11.0 , tell the same story. The contrarian alpha score showed nothing (Spearman $+0.01$, $p = 0.71$), and nor did the options direction: calls minus puts, $+2.8$ percentile points ($p = 0.54$). Puts were right 64% of the time, but only because most stocks trailed SPY that year; only 44% of the calls beat it.

The clean tests do not agree

Tests the model could not pass from memory / percentile points of excess return



Sources: evaluation/results.json; evaluation/origin-ledger/results.json. One year unless stated.

Figure 9: The clean tests do not agree. Each point is a high-minus-low difference in mean percentile rank of excess return; grey bands hold 95% of within-industry shuffles. Only the EON verdict was read after its outcome window had begun, from filings the model could not have seen; the three 2025 scores were recorded before any outcome existed. The EON statistic was chosen after seeing the tails (§8). About 37% of 2025 ledger companies have no industry label and share one block. Sources: evaluation/results.json; evaluation/origin-ledger/results.json.

These results do not contradict the EON verdicts: across 1,054 companies in both sets, the verdict and the compounder score are almost unrelated (Spearman 0.05). They are different judgements. They do share a market: May 2025 to May 2026, a year in which the median stock in the ledger trailed SPY by more than ten points.

9.6 Conviction adds little

The February report found that high-conviction calls roughly doubled the spread. With better parsing and the full panel, it does not. At one year, 653 high-conviction BUY and 729 high-

conviction SELL readings give 13.3 points, against 11.9 for all readings. Conviction labels are real but add little information beyond the verdict.

9.7 Reading the results together

Two explanations fit the historical result. The model may have known how those years went. Or its lenses, which favour strong balance sheets and durable businesses, may have suited 2021 to 2024, when such companies did well. The historical data cannot separate them. They are not mutually exclusive.

The post-cutoff data add one observation to that argument. The industry tilt, the part most easily explained by hindsight about sectors, disappears after the cutoff. The within-industry spread, the part that looks most like reading a filing carefully, mostly stays. That is the pattern skill would produce. It rests on one vintage, one regime and a statistic chosen after seeing the tails.

Across every test the model could not have passed from memory, one points up, one points down and two point nowhere (Figure 9).

Before the cutoff, EON's verdicts sorted winners from losers too well to trust. After it, one verdict still sorts them and another sorted them backwards. A fluent reader is not thereby a forecaster.

10. What the work taught me

10.1 Fix the question before measuring the answer

The workflow builder produced better individual analyses than the market-wide batch. It produced no evidence. Only a fixed schema, applied to everything, made the model's judgement something that could be counted, compared and tested.

10.2 Engineer for the scarcest resource

For EON the scarce resource was requests, not compute or storage. That made per-year checkpoints, leases, resumption and quota-aware waiting the core of the system. It also made call count a design variable: one call for a wide screener, four for a framework too large for one schema.

10.3 Let the model judge; let code count

Two lessons point the same way. The February backtest found that the model's categorical verdict carried more signal than an additive composite of its own sub-scores. The CSPP rewrite found that the model's arithmetic drifts. Ask the model for judgements on anchored scales, and do the adding in code, where it can be audited and reweighted.

10.4 Date everything the model sees and says

The most useful field in the database turned out to be `filing_date`. It made entry dates exact and made the cutoff test possible. Every reading should carry the model's identity and cutoff alongside the filing's date.

10.5 An underpowered null is not a negative result

The first run, on prices ending in February 2026, found a post-cutoff six-month spread of +1.0 points, and -0.1 within industry. It was tempting to report that the effect vanished. The test could only have detected about 8 to 10 points. With more filings observed, the same test gives +4.6. “Could not detect” is a different finding from “absent”, and the minimum detectable size belongs next to every null.

10.6 Look at the tails before trusting a mean

A single stock moved the post-cutoff one-year mean by about 10 points. Returns of small companies are heavy-tailed, and a mean over a few hundred of them can belong to one. A rank statistic or a median should sit beside every mean.

10.7 Resemblance is not a forecast

The compounder score measured something real: how closely a company's filings resembled those of proven compounders. It was a sensible idea and a confident model applied it carefully. For the year that followed, it pointed the wrong way. A plausible reason, untested here, is that resemblance to past winners was already priced. The lesson is simpler: a score is a hypothesis until a ledger says otherwise.

10.8 Tell the model the date

A model reasons from its training-time present unless told otherwise. In 2025 it dated its work a year early and proposed options that had already expired. EON's later workflows stamp run metadata in code; the date belongs in the prompt too.

11. Scope and remaining questions

The historical result cannot separate memory from judgement. Every outcome before the cutoff may be in the model's training data. The anachronism probe rules out careless leakage only.

Every clean test shares one market. The post-cutoff EON sample and the 2025 ledger both run through 2025 and 2026. A result that holds in one regime may reverse in the next, as the compounder score may show.

The clean evidence is one vintage. Nearly all post-cutoff filings were published in the first quarter of 2025 and held through one market: the tariff shock of April 2025 and the rally that followed. One regime cannot show that a result generalises.

The strongest clean result used a statistic chosen late. The rank statistic was added after inspecting the tails. It was applied identically to every sample and has no free parameters, but it was not pre-registered, and the pre-registered six-month test gave $p = 0.11$ and 0.048 .

Industry is a control, not a factor model. Size, value, profitability and momentum are not controlled. A lens that prefers strong balance sheets is, partly, a quality factor. Point-in-time fundamentals would be needed to separate them.

The universe is survivors. Companies were chosen from 2026 listings, and 45 companies acquired or taken private by September 2026 have no refreshed prices. For the 2025 ledger, 120 of 710 additional tickers returned no prices. The direction of this bias for the spread is not known.

One model, one prompt. All readings come from Gemini 2.5 Flash and one prompt. Other models, prompts, or the same model at another temperature may disagree.

Prices come from one source. Yahoo Finance adjusted closes, checked for coverage but not against a second vendor.

Returns are not trades. No transaction costs, borrow costs, position sizing or capacity are modelled. The spread is a measure of information, not of profit.

12. Future work

12.1 Keep the ledger

The 2025 scripts showed what a ledger is worth: it needs no cutoff argument at all. EON should keep one deliberately. The fiscal 2025 10-Ks filed in early 2026 are not yet in the database, because the batch ran before most were filed. Reading them now with the same model and prompt, and freezing each verdict with its date, would roughly double the clean sample by spring 2027.

12.2 Freeze the model

The clean window belongs to one model. Switching to a model with a later cutoff moves the line and turns part of the clean sample back into memory. The evaluation model should be pinned, and a new model should start its own ledger. Models are retired, so this has a lifespan; the pinned version and its cutoff should be recorded with every reading.

12.3 Read filings anonymously

Remove the company's name, ticker, and identifying product names from a sample of historical filings and read them again. If the historical spread shrinks under anonymity, memory was doing part of the work. Glasserman and Lin used this design for news headlines.

12.4 Control for known factors

With point-in-time market capitalisation, valuation and profitability, the spread can be tested within size and quality groups, or against a factor model. That would say whether the verdicts know anything a quality screen does not.

12.5 Hold the hypothesis workflows to the same standard

The Moonshot $\times$ Excellence and CSPP workflows have produced few readings. Before their scores are trusted, each needs the same treatment: a fixed protocol, dated readings, and a clean sample after the cutoff.

13. Related work

Look-ahead bias in language models. Sarkar and Vafa show that pretrained language models leak future information into tasks meant to use only the past, and test for it with events that should be unpredictable. The anachronism probe here is a simple instance of that idea. Sarkar and Vafa, Lookahead Bias in Pretrained Language Models. Glasserman and Lin measure look-ahead bias in GPT sentiment scores for news headlines, and remove company identifiers to reduce it. Glasserman and Lin, Assessing Look-Ahead Bias in Stock Return Predictions Generated by GPT Sentiment Analysis.

Language models and returns. Lopez-Lira and Tang find that ChatGPT's reading of news headlines predicts next-day returns, and pay close attention to the model's training window. Lopez-Lira and Tang, Can ChatGPT Forecast Stock Price Movements?. EON differs in its input (whole annual reports), its horizon (months to years) and its output (a verdict from three stated philosophies).

Reading 10-Ks. Textual analysis of annual reports has a long history. Loughran and McDonald built finance-specific word lists for 10-Ks, and Cohen, Malloy and Nguyen found that changes in 10-K text predict returns. These methods count words; EON asks a model for a judgement. Both face the same question of what the text knows that prices do not. Loughran and McDonald, When Is a Liability Not a Liability?; Cohen, Malloy and Nguyen, Lazy Prices.

Multiple testing in finance. Harvey, Liu and Zhu show how many published return predictors fail once the number of tests is counted. That argument is why the primary tests here were fixed first and later additions are labelled. Harvey, Liu and Zhu, ... and the Cross-Section of Expected Returns.

14. Conclusion

I set out to build a reader that gives every annual report the same careful attention. It began in May 2025 as scripts studying excellent companies. The workflow builder showed which questions were worth asking. Fixing one question and asking it of every filing turned the model into an instrument. The quota set the pace, and 6,653 readings later there was something to measure.

The first measurement was too good. It tested the model on years it may already have known. Rebuilt around the model's knowledge cutoff, the test tells a quieter story. Before the cutoff, the verdicts separate winners from losers by 13 points a year, and part of that is a bet on industries. After it, the industry bet is gone, and a smaller within-industry spread remains. And the scores written down in 2025, before anyone knew the answers, include one that pointed the wrong way.

That is where the evidence stands today. It will not stay there: every filing season adds readings the model cannot have memorised. The newspaper the model read ends in January 2025. Everything after that has to be read the ordinary way, one day at a time.

Appendix A: reproduction

Artefacts

Path	Contents
eon/ai/prompts/perspectives.py	The three-lens prompt
eon/analysis/perspectives/analyzer.py	Filing text to structured reading
eon/ai/providers/gemini.py	Model calls, schema validation, retries
eon/ui/services/batch_queue.py	Multi-day batches, leases, resumption
custom_workflows/cspp_v26_analyzer.py	Four-call split, totals in code
custom_workflows/moonshot_excellence_finder.py	One-call screener, anchored scales
experimental/backtester/BACKTEST_REPORT.md	The February report (§7.1)
docs/whitepaper/evaluation-config.yaml	The protocol and its amendments
docs/whitepaper/origin-ledger-config.yaml	The 2025 ledger protocol
stock_stuff_06042025/10K_automator/	The 2025 origin scripts and outputs (read-only)

Rebuilding the evaluation, figures and paper

From the eon root. The database is opened read-only; outputs go only to `docs/whitepaper/`. The price refresh needs network access and `yfinance`; every other step runs offline. Evaluation and figures need NumPy, pandas, SciPy, Matplotlib and PyYAML.

```
python docs/whitepaper/refresh_prices.py          # optional: fetch new prices
python docs/whitepaper/evaluate_backtest.py       # about 30 seconds
python docs/whitepaper/refresh_prices.py --origin # optional: 2025 ledger tickers
python docs/whitepaper/evaluate_origin_ledger.py
python docs/whitepaper/prepare_figure_data.py
python docs/whitepaper/render_figures.py
python docs/whitepaper/build_paper.py             # needs Pandoc 3.x
```

A refresh moves `price_end` only if the config is changed. Every script writes a `run.json` with its configuration and the hashes of its inputs and outputs. The refreshed price file is local and not committed; its hash is recorded.

Appendix B: glossary

Reading. One model call on one 10-K, stored as 35 validated fields.

Vintage. All readings with the same fiscal-year label.

Verdict. The first word of the final verdict: STRONG BUY, BUY, HOLD, SELL or STRONG SELL. BUY and STRONG BUY form the long leg; SELL and STRONG SELL the short.

Excess return. A stock's return minus SPY's over the same window, from the first trading day after the filing.

Spread. Mean excess return of the long leg minus that of the short leg, in percentage points.

Rank spread. The same difference, using each reading's percentile rank of excess return within its vintage. One extreme return cannot decide it.

Knowledge cutoff. The date after which a model's training data is said to end. For Gemini 2.5 Flash, January 2025; this paper uses 31 January 2025.

Industry tilt. The mean of the within-industry permutation null: the spread expected from the model's choice of industries alone.

Minimum detectable spread. About 2.8 times the permutation null's standard deviation: the spread a test would detect with 80% power at $p < 0.05$.

Anachronism. A reading that mentions something that did not exist when its filing was published.

Appendix C: figures at a glance

ORIGIN, MAY-JUNE 2025	11,741 filing readings, 2,175 companies scores recorded before their outcomes
READINGS	6,653 company-years, 1,358 companies
MODEL	Gemini 2.5 Flash, cutoff January 2025
SCHEMA	35 fields: 3 lenses x 11, synthesis, verdict
MARKET BATCH	6,568 readings in 13 days, 8-21 Feb 2026
QUOTA	25 keys x 20 requests = 500 readings a day
VERDICT MIX	HOLD 50.8% BUY 27.5% SELL 16.5% STRONG BUY 3.8% STRONG SELL 1.4%
FEBRUARY REPORT	2Y spread +14.1 pts (reported; pre-cutoff)
BEFORE CUTOFF, 1Y rank spread	+13.1 pts (1,099 BUY / 657 SELL) p = 0.0001 11.5: 4.8 industry tilt, 6.6 within industry
AFTER CUTOFF, 6M	+4.6 pts (416 / 275) p = 0.11; 0.048 by industry
AFTER CUTOFF, 1Y mean one stock (BW)	+14.9 pts (339 / 201) +/-23; p = 0.22 adds about 10 pts to the BUY mean
AFTER CUTOFF, 1Y rank	5.9: 0.1 industry tilt, 5.9 within p = 0.0051 (rank statistic added after seeing the tails)
2025 LEDGER, 1Y	compounder: Spearman -0.14 p = 0.0001 contrarian alpha: +0.01 p = 0.71 options direction (6M): +2.8 p = 0.54
ANACHRONISM PROBE	0 early mentions; IRA in 300 readings
HIGH CONVICTION, 1Y	+13.3 pts vs +11.9 for all readings
NOT YET MEASURED	2Y after cutoff; anonymised reading; a second regime; factor controls; a second clean vintage

Prepared from the EON repository, its analysis database, and a price refresh of 29 September 2026. Figures are drawn from a checked-in evidence snapshot with source hashes. Negative and inconclusive results are retained. This paper is research, not investment advice. See figures.md for figure provenance and revision-notes.md for what needs the author's confirmation.